

Guided-TTS: A Diffusion Model for Text-to-Speech via Classifier Guidance

Converting Text to Speech: A foray into TTS models

Neural Text-to-Speech (TTS) models are able to generate high-quality human-like speech from given text. TTS models are generally conditional generative models that encode text into a hidden representation and generate speech from the encoded representation. Earlier models were generally “autoregressive”, meaning that the current value is a function of the past values. They suffered from a slow synthesis speed due to the sequential nature of the sampling procedure. Today, owing to the great advancements in this field, we are able to generate raw waveforms at a very fast speed.

Despite the high quality and fast inference speed, most TTS models can only be trained if the transcribed data of the target speaker are provided. Untranscribed data is easily available in the form of audiobooks and podcasts, but utilizing it for existing TTS models remains a challenge due to the conditional modelling of speech distribution. There have also been approaches using untranscribed speech to adapt pre-trained multi-speaker models for few-shot TTS synthesis. The issue is that these adaptive TTS models rely on pre-training multi-speaker TTS models, which are challenging to train and require high data quality. These models do not generalize well and underperform in comparison to high-quality single-speaker TTS models.

Make way for Guided-TTS!

The paper proposes Guided-TTS, a high quality TTS model that learns to generate speech with an unconditional DDPM and performs text-to-speech synthesis using classifier guidance. It uses a phoneme classifier trained on existing large-scale ASR datasets, and does not use any transcript of the target speaker for TTS. The model also learns to generate mel-spectrograms without context. As the untranscribed data does not have to be aligned with the text sequence, random chunks of untranscribed speech are used for training. This allows the building of training datasets without any extra effort in modelling the speech of speakers for whom no transcribed data is available.

To guide the unconditional DDPM for TTS, a frame-wise phoneme classifier is trained on the LibriSpeech dataset, and the gradient of the classifier is used during sampling. In order to avoid mispronunciation through guiding error, which could be fatal for the TTS model, norm-based guidance is presented.

Denoising Diffusion Probabilistic Models (DDPM)

DDPMs, which are a type of probabilistic generative model, have been recently applied to images and audio both. DDPM first defines a forward process that gradually corrupts data X_0 into random noise X_T across T timesteps. The model is trained to learn the reverse process, which follows the reverse trajectory of the predefined forward process to generate data from random noise.

Recent approaches propose formulating the trajectory between data and noise as continuous stochastic differential equation (SDE) instead of using a discrete-time Markov process, as done in the original paper. Guided-TTS uses Grad-TTS, which introduces SDE formulation to TTS. According to Grad-TTS, the forward corruption process from X_0 to X_T is given by

$$dX_t = -\frac{1}{2}X_t\beta_t dt + \sqrt{\beta_t}dW_t$$

Here, β_t is a predefined noise schedule given by

$$\beta_t = \beta_0 + (\beta_T - \beta_0)t$$

and W_t is a Wiener process. As shown by Anderson (1982), the reverse process, which represents a trajectory from X_T to X_0 , can also be formulated as an SDE, defined as follows

$$dX_t = \left(-\frac{1}{2}X_t - \nabla_{X_t} \log p_t(X_t)\right)\beta_t dt + \sqrt{\beta_t}d\widetilde{W}_t$$

Here, the last term represents a reverse-time Wiener process. Given the **score**, which is the gradient of the log-density with respect to the data, we sample data X_0 from random noise X_T , by solving the above reverse process SDE. To generate data, the DDPM learns to estimate the score by using a parametrized neural network. In order to estimate the score, X_t has to be sampled from the distribution given by the forward SDE and the data X_0 , which is as follows

$$\begin{aligned} X_t|X_0 &\sim \mathcal{N}(\rho(X_0, t), \lambda(t)) \\ \rho(X_0, t) &= \exp\left(-\frac{1}{2}\int_0^t \beta_s ds\right) \times X_0 \\ \lambda(t) &= I - \exp\left(-\int_0^t \beta_s ds\right) \end{aligned}$$

The score can then be derived from the following equation

$$\nabla_{X_t} \log p_t(X_t|X_0) = \lambda(t)^{-1} \varepsilon_t$$

ε_t : standard Gaussian noise used to sample X_t given X_0

The training for the model for all time-steps can be done using the following L2 loss function given by

model: $s_\theta(X_t, t)$ for $\forall t \in [0, T]$

$$\text{Loss: } L(\theta) = \mathbb{E}_t \mathbb{E}_{X_0} \mathbb{E}_{\varepsilon_t} [\|s_\theta(X_t, t) + \lambda(t)^{-1} \varepsilon_t\|_2^2]$$

Using this model, we can generate samples X_0 from noise X_T . In case of Grad-TTS, we set $T = 1$ and use a fixed discretization strategy

$$X_{t-\frac{1}{N}} = X_t + \frac{\beta_t}{N} \left(\frac{1}{2} X_t + \nabla_{X_t} \log p_t(X_t) \right) + \sqrt{\frac{\beta_t}{N}} z_t$$

N is the number of steps required to solve the SDE

$$t \in \left\{ \frac{1}{N}, \frac{2}{N}, \dots, 1 \right\}$$

z_i is standard Gaussian noise

Classifier Guidance

DDPM can be guided to generate samples with the desired condition without fine-tuning through the introduction of a classifier. Unconditional DDPM can be used to generate class-conditional images by applying a separately trained image classifier. For conditional generation, the classifier can be trained to classify noisy data in a conditional way.

We can replace the unconditional score from the earlier equations by the conditional score

Classifier $p_t(y|X_t)$ is trained to classify noisy data X_t as condition y

$$X_{t-\frac{1}{N}} = X_t + \frac{\beta_t}{N} \left(\frac{1}{2} X_t + \nabla_{X_t} \log p_t(X_t|y) \right) + \sqrt{\frac{\beta_t}{N}} z_t$$

$$\nabla_{X_t} \log p_t(X_t|y) = \nabla_{X_t} \log p_t(X_t) + \nabla_{X_t} \log p_t(y|X_t)$$

The above equations allow sampling X_0 with condition y if the unconditional score and classifier gradient for the target condition are given. An additional improvement is to introduce a gradient scale 's' when guiding the DDPM, which is multiplied to the classifier gradient to adjust the scale of it. Using $s > 1$ generates high-fidelity (but less diverse) samples, which contributes to achieving SOTA performance for class-conditional generation.

Guided-TTS

Guided-TTS is a high-quality text-to-speech model which can be trained without any transcript of the target speaker. It learns to model unconditional distribution of speech and generates speech from text using classifier guidance. For classifier guidance, an unconditional diffusion model is trained on untranscribed speech data and a phoneme classifier trained on a large-scale speech recognition dataset.

It consists of four modules:

1. Unconditional DDPM
2. Phoneme classifier
3. Duration predictor
4. Speaker encoder

Unconditional DDPM

It models the unconditional distribution of speech without any transcript. Speech data from a single target speaker is used to train the diffusion model to build a TTS for that speaker. Random chunks of speech can be used for training since we don't need to align it with the transcriptions. Given a mel-spectrogram, the forward process corrupts the data into noise, and the reverse process estimates the unconditional log likelihood score for each timestep, as shown in the previous section. At each iteration, X_t , t is sampled from the mel-spectrogram X_0 , and the score is estimated using the neural network "s". We regard mel-spectrogram as a 2D image with a single channel and use U-net architecture as "s".

Text-to-Speech via Classifier Guidance

Using classifier guidance decouples the generative modeling of speech by conditioning text information, which allows the use of noisy speech recognition dataset for the phoneme classifier, which is challenging for the existing TTS models

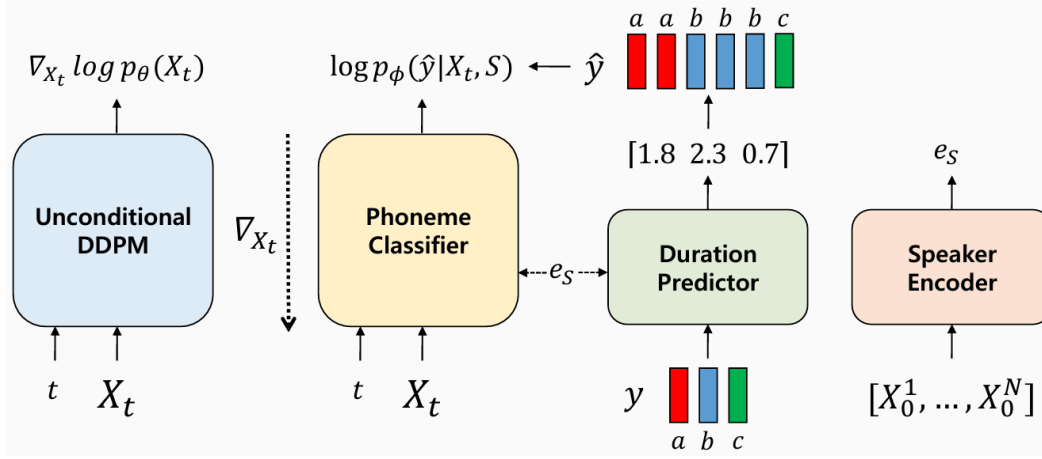


Figure 1: The overall architecture of Guided-TTS. The unconditional DDPM learns to generate speech X_0 with untranscribed data. The other modules, the phoneme classifier, duration predictor, and speaker encoder are for guiding the unconditional DDPM to generate conditional samples given y .

As shown in the figure, to generate mel-spectrogram given text, we expand the transcript y to frame-level phoneme label y' , using the duration predictor. We then sample a random noise X_t of the same length as y' from the standard normal distribution, to generate conditional samples with a conditional score.

We build a TTS model with the unconditional generative model by adding the gradient of the phoneme classifier during the generative process.

$$\begin{aligned} \nabla_{X_t} \log p(X_t | \hat{y}, \text{spk} = S) &= \nabla_{X_t} \log p_\theta(X_t | \text{spk} = S) \\ &\quad + \nabla_{X_t} \log p_\phi(\hat{y} | X_t, \text{spk} = S) \end{aligned} \quad (8)$$

The phoneme classifier and duration predictor are trained on a large-scale speech recognition dataset for better generalization to the unseen speaker S .

Phoneme Classifier

A WaveNet like architecture is trained to recognize the phoneme corresponding to each frame of the input mel-spectrogram. For training the frame-wise phoneme classifier, we align the transcript and speech using Montreal Forced Aligner (MFA), and extract frame-level phoneme label y' . The phoneme classifier is used to train the corrupted mel-spectrogram X_t . The training objective here is to minimize the cross-entropy between the phoneme label y' and the output probability with respect to t .

Duration predictor

This module predicts the duration of each text token for a given text sequence y . The objective here is to minimize the L2 loss between the duration label and the estimated duration in the log-domain. The architecture of this module is taken from Glow-TTS. Text embeddings and speaker embeddings are concatenated to predict the speaker-dependent duration

Speaker Encoder

This module encodes the speaker information from mel-spectrogram and outputs the speaker embedding. A speaker encoder is trained using GE2E loss on the speaker verification dataset. Speaker embeddings are extracted from the clean mel-spectrogram X_0 for each training data. For guidance, we average and normalise the speaker embeddings of the untranscribed speech for the target speaker S .

Algorithm 1 Norm-based Guidance

$\hat{y}$: frame-wise phoneme label, s : gradient scale, τ : temperature
 θ : parameter of unconditional DDPM, ϕ : parameter of phoneme classifier
 $X_1 \sim \mathcal{N}(0, \tau^{-1}I)$
for $i = N$ **to** 1 **do**
 $t \leftarrow \frac{i}{N}$
 $\alpha_t \leftarrow \|\nabla_{X_t} \log p_\theta(X_t)\| / \|\nabla_{X_t} \log p_\phi(\hat{y}|X_t)\|$
 $z_t \sim \mathcal{N}(0, \tau^{-1}I)$
 $\mu_t \leftarrow \frac{1}{2}X_t + \nabla_{X_t} \log p_\theta(X_t) + s \cdot \alpha_t \nabla_{X_t} \log p_\phi(\hat{y}|X_t)$
 $X_{t-\frac{1}{N}} \leftarrow X_t + \frac{\beta_t}{N}\mu_t + \sqrt{\frac{\beta_t}{N}}z_t$
end for
return X_0

When guiding the unconditional DDPM with the frame-wise phoneme classifier, it was found that the norm of the unconditional score suddenly increases near $t=0$. That is, when closer to data X_0 , the phoneme classifier has little effect on the generative process of the DDPM. Therefore, a norm-based guidance is proposed which scales the norm of the classifier gradient in proportion to the norm of the score in order to prevent the effect of the gradient from being insignificant as the score steeply increases. By adjusting the gradient scale, we can determine how much the classifier gradient contributes to the guidance of Unconditional DDPM. It was also observed that tuning the temperature parameter to a value greater than 1 generates high-quality mel-spectrograms.